

A proper-infiniteness obstruction to finite-dimensional Ext^1

Partial result packet for arXiv:2006.07696

Run: fa_banach_001

Source and target question

The source paper is P. Kuchment, *Three-representation problem in Banach spaces*, arXiv:2006.07696 [1]. In Section 4, Kuchment asks:

4. CAN $\text{Ext}^1(F, E)$ BE OF A POSITIVE FINITE DIMENSION?

We do not know¹¹ whether there exist (infinite dimensional) Banach spaces E and F such that $0 < \dim(\text{Ext}^1(F, E)) < \infty$. We describe below some results (also stated without a proof in [34]) that provide several classes of spaces where such possibility is excluded.

In text form, the question is whether there are infinite-dimensional Banach spaces E and F such that

$$0 < \dim \text{Ext}^1(F, E) < \infty.$$

The source proves several zero-or-infinite criteria. One of them is [1, Theorem 4.6]: if $E \oplus E \cong E$, or if $F \oplus F \cong F$, then $\text{Ext}^1(F, E)$ is either zero or infinite-dimensional. The proof uses the absence of nonzero finite-dimensional unital representations of the corresponding operator algebra.

Result

Definition 1. Let X be a Banach space. Say that the identity of $\mathcal{L}(X)$ is properly infinite if there are operators $U_1, U_2, V_1, V_2 \in \mathcal{L}(X)$ such that

$$V_i U_i = I_X \quad (i = 1, 2), \quad (U_1 V_1)(U_2 V_2) = 0.$$

Equivalently, X contains two complemented subspaces X_1, X_2 , each isomorphic to X , with the corresponding projections orthogonal.

Theorem 1 (Properly infinite obstruction). Let E and F be Banach spaces. If the identity of $\mathcal{L}(E)$ is properly infinite, or if the identity of $\mathcal{L}(F)$ is properly infinite, then

$$\text{Ext}^1(F, E) = 0 \quad \text{or} \quad \dim \text{Ext}^1(F, E) = \infty.$$

Corollary 1. The conclusion holds whenever E , or F , contains two orthogonal complemented copies of itself. In particular, Kuchment's case $E \oplus E \cong E$ or $F \oplus F \cong F$ is a special case, but the two self-copies need not sum to the whole space.

Proof intuition

Kuchment proves that $\mathcal{L}(E)$ acts on $\text{Ext}^1(F, E)$, and that $\mathcal{L}(F)$ acts on the right, equivalently by an anti-representation. These actions are unital: the identity operator acts as the identity on $\text{Ext}^1(F, E)$.

Suppose $\text{Ext}^1(F, E)$ were nonzero and finite-dimensional. If $P = UV$ is an idempotent equivalent to the identity, with $VU = I$, then the image of U has a one-sided inverse in a finite-dimensional algebra of operators. Hence it is invertible, and the image of P must be the identity. Two orthogonal such idempotents would both map to the identity, while their product maps to zero. This is impossible.

Proof

Assume first that the identity of $\mathcal{L}(E)$ is properly infinite. Let

$$V = \text{Ext}^1(F, E)$$

and suppose toward contradiction that $0 < \dim V < \infty$. By [1, Theorem 1.8], $\mathcal{L}(E)$ acts on V by a unital representation. Write the action of $A \in \mathcal{L}(E)$ as $\widetilde{A}$.

Let $U_i, V_i \in \mathcal{L}(E)$, $i = 1, 2$, witness proper infiniteness:

$$V_i U_i = I_E, \quad P_i := U_i V_i, \quad P_1 P_2 = 0.$$

Then

$$\widetilde{V}_i \widetilde{U}_i = \widetilde{I}_E = I_V.$$

Since V is finite-dimensional, a linear operator with a left inverse is invertible, and its left inverse is also its right inverse. Thus

$$\widetilde{P}_i = \widetilde{U}_i \widetilde{V}_i = \widetilde{U}_i \widetilde{V}_i = I_V \quad (i = 1, 2).$$

But $P_1 P_2 = 0$, so

$$0 = \widetilde{0} = \widetilde{P_1 P_2} = \widetilde{P_1} \widetilde{P_2} = I_V,$$

contradicting $V \neq 0$. Hence no nonzero finite-dimensional $\text{Ext}^1(F, E)$ can occur.

Now assume instead that the identity of $\mathcal{L}(F)$ is properly infinite. By [1, Theorem 1.8], $\mathcal{L}(F)$ acts on $V = \text{Ext}^1(F, E)$ on the right; denote this anti-representation by R_A , so that

$$R_{AB} = R_B R_A, \quad R_{I_F} = I_V.$$

Choose $U_i, V_i \in \mathcal{L}(F)$ with $V_i U_i = I_F$ and $(U_1 V_1)(U_2 V_2) = 0$. From $V_i U_i = I_F$ we get

$$R_{U_i} R_{V_i} = I_V.$$

In finite dimension this again implies $R_{V_i} R_{U_i} = I_V$. Therefore, for $P_i = U_i V_i$,

$$R_{P_i} = R_{U_i V_i} = R_{V_i} R_{U_i} = I_V.$$

Since $P_1 P_2 = 0$,

$$0 = R_0 = R_{P_1 P_2} = R_{P_2} R_{P_1} = I_V,$$

again a contradiction. This proves the theorem.

Verification notes and limitations

The argument is a direct strengthening of Kuchment's Lemma 4.5. The source uses $E \oplus E \cong E$ to obtain two complementary self-copies summing to the identity. The proof above only uses the algebraic consequences

$$V_i U_i = I, \quad (U_1 V_1)(U_2 V_2) = 0,$$

and therefore applies to any properly infinite identity in the operator algebra.

This does not solve the full finite positive-dimensional Ext^1 problem. The residual candidate spaces are precisely those where this kind of finite-dimensional representation obstruction may fail, such as Dedekind-finite/few-operators spaces. The prior attempt for this target identified the same frontier.

References

- [1] P. Kuchment, *Three-representation problem in Banach spaces*, arXiv:2006.07696, 2020. <https://arxiv.org/abs/2006.07696>